AdventureWorks Financial Analysis - Project Log

1. Project Objective

Analyze AdventureWorks financial and sales performance to identify the products, categories, customers, and regions that drive revenue and profitability, evaluate sales performance against quotas, uncover growth opportunities and inefficiencies, and provide data-driven recommendations for improving overall business performance.

2. Dataset Overview

- Dataset: AdventureWorks DW 2008
- Source: Kaggle
- Type: Data Warehouse
- Main fact tables:
    - FactInternetSales
    - FactFinance
    - FactSalesQuota
- Main dimension tables:
    - DimProduct
    - DimProductCategory
    - DimProductSubcategory
    - DimCustomer
    - DimGeography
    - DimSalesTerritory
    - DimDate

3. Data Model Validation

Goals

- Identify primary keys
- Identify foreign keys
- Validate table relationships
- Build star schema

Tables Imported

- FactInternetSales
- FactFinance
- FactSalesQuota
- DimProduct
- DimProductCategory
- DimProductSubcategory
- DimCustomer
- DimGeography
- DimSalesTerritory
- DimDate
- DimScenario

Primary Keys Identified

- DimProduct → ProductKey
- DimCustomer → CustomerKey
- DimDate → DateKey
- DimGeography → GeographyKey
- DimSalesTerritory → SalesTerritoryKey
- DimProductCategory → ProductCategoryKey
- DimProductSubcategory → ProductSubcategoryKey
- DimEmployee → EmployeeKey
- DimScenario → ScenarioKey

Notes:

Some dimension tables contain alternate keys in addition to primary keys. For example, DimProduct contains ProductAlternateKey and DimCustomer contains CustomerAlternateKey. These appear to be business-readable identifiers, while the numeric primary keys are used to establish relationships and perform joins between tables.

FactInternetSales contains foreign keys that connect the sales transactions to the dimension tables:

- ProductKey connects to DimProduct
- CustomerKey connects to DimCustomer
- SalesTerritoryKey connects to DimSalesTerritory
- OrderDateKey connects to DimDate

These relationships allow sales, cost, and profit measures to be analyzed by product, customer, region, and time.

Financial Planning Structure Validated

FactFinance is linked to DimScenario through ScenarioKey.

DimScenario contains three financial planning scenarios:

- 1 = Actual
- 2 = Budget
- 3 = Forecast

This enables comparisons between actual financial performance, planned budgets, and forecasted values. As a result, budget variance and forecasting analyses may be incorporated into the project.

4. Business Questions

Time & Growth Analysis

1. How has total revenue and profit trended year-over-year?
2. Which months or quarters consistently drive peak and low sales?
3. How does revenue growth vary by region year-over-year?

Product Performance

4. Which products generate the highest revenue?
5. Which products generate the highest profit margin?
6. Which products generate high revenue but low margins?
7. Which categories drive the most profitability and which drag it down?

Regional Performance

8. Which regions generate the highest revenue and profit?
9. Which regions underperform relative to their costs?
10. Which regions have the highest revenue per customer and profit per order?
11. Which product categories perform differently across regions?

Customer Analysis

12. Which customer occupation segments generate the highest revenue and profit per customer?

13. What is the average order value and purchase frequency by customer segment?
14. Which demographic customer segments are the most and least profitable?
15. What factors are most strongly associated with high-value customers?

Sales Performance

16. How do actual sales compare with sales quotas by region and salesperson?
17. Which salespeople or territories consistently miss quota and by how much?
18. What is the quota attainment rate across the organization over time?

    Note: Sales performance analysis was excluded from the project due to insufficient data. The available quota data covered only a limited number of employees and regions, making the results unsuitable for reliable performance evaluation and comparison.

Strategic Recommendations

19. Which products and categories should be prioritized to maximize revenue growth and profit generation?
20. How can the company improve profit margins without sacrificing revenue volume?
21. Which regions should be prioritized for targeted growth investment, and which require cost scrutiny?
22. How concentrated is the company's revenue and profit risk, and what steps should management take to reduce exposure?
23. Which customer segments should receive the most marketing and retention investment, and which are lower priority?
24. What resource reallocation across product categories would most improve overall business profitability?
25. How should the company adjust its growth strategy given the pattern of rising volume but declining average order value?
26. Which regional markets contain the highest-value customers, and how should marketing budgets be allocated across them?

5.Data Cleaning & Quality Assessment

Check 0: Table Import Validation & Row Counts

Objective

Verify that all selected tables were imported successfully into SQLite and contain data required for analysis.

Tables Checked

- FactInternetSales
- DimProduct
- DimCustomer
- DimDate
- DimGeography
- DimSalesTerritory
- DimProductSubcategory
- DimProductCategory
- FactSalesQuota
- DimEmployee
- FactResellerSales

Result

Table	Row Count
FactInternetSales	60,398
DimCustomer	18,484
DimDate	1,188
DimGeography	655
DimProduct	606
DimProductSubcategory	37
DimProductCategory	4
DimSalesTerritory	11
FactSalesQuota	163
DimEmployee	296
FactResellerSales	60,855

Conclusion

- All selected tables imported successfully.

- No tables were empty.
- FactInternetSales contains sufficient transaction volume for meaningful analysis.
- Supporting dimension tables were successfully loaded and are available for joins and data modeling

Check 1: Critical Null Values

Objective
Verify that essential analytical fields are populated.

Columns Checked

- ProductKey
- CustomerKey
- SalesTerritoryKey
- OrderDateKey
- SalesAmount
- TotalProductCost

Result

- No null values found in any critical analytical fields.

Conclusion
Dataset passed critical null-value validation and required no remediation.

Check 2: Duplicate Sales Transactions

Objective
Verify transaction-level uniqueness.

Method
Checked duplicate combinations of:

- SalesOrderNumber
- SalesOrderLineNumber

Result

- Query returned 0 rows.

- No duplicate transactions found.

Conclusion
FactInternetSales passed duplicate integrity validation.

Check 3: Suspicious Financial Values

Objective
Identify invalid transactions that could distort profitability analysis.

Checks

- SalesAmount < 0
- TotalProductCost < 0
- OrderQuantity ≤ 0

Result

- 0 suspicious rows identified.

Conclusion
No invalid sales, cost, or quantity values detected.

Check 4: Date Coverage Validation

Conclusion

The dataset contains **37 months of sales history**.

The available time period is sufficient for year-over-year, quarterly, monthly, product, regional, customer, and sales performance trend analysis.

No invalid or unexpected date values were identified.

Check 5: Referential Integrity / Orphan Key Check

Relationships Checked

- FactInternetSales.ProductKey → DimProduct.ProductKey

- FactInternetSales.CustomerKey → DimCustomer.CustomerKey
- FactInternetSales.SalesTerritoryKey → DimSalesTerritory.SalesTerritoryKey
- FactInternetSales.OrderDateKey → DimDate.DateKey
- FactResellerSales.EmployeeKey → DimEmployee.EmployeeKey
- FactResellerSales.SalesTerritoryKey → DimSalesTerritory.SalesTerritoryKey
- FactSalesQuota.EmployeeKey → DimEmployee.EmployeeKey
- FactSalesQuota.DateKey → DimDate.DateKey

Result

- Orphan products: 0
- Orphan customers: 0
- Orphan territories: 0
- Orphan dates: 0

Conclusion

All core fact-to-dimension relationships are valid. The sales fact table can be safely joined with product, customer, territory, and date dimensions for analysis in Python and Power BI.

Check 6: Validate Sales Quota Completeness

Objective: Ensure quota targets are available for all records and can be used for sales performance analysis.

Columns Checked: SalesAmountQuota

Result:

- No missing quota values found (0 NULLs).
- Sales quota data is complete and suitable for quota attainment analysis.
- FactSalesQuota confirmed as a reliable source for performance benchmarking.

Data Cleaning & Quality Assessment Summary

Objective

Assess data quality and ensure the dataset is reliable for financial and business analysis.

Findings

Check	Result
Row Count Verification	All tables imported successfully and row counts were validated.
Missing Values in Key Columns	No null values found in ProductKey, CustomerKey, SalesTerritoryKey, OrderDateKey, SalesAmount, or TotalProductCost.
Duplicate Transaction Check	No duplicate sales transactions detected.
Invalid Value Check	No negative sales amounts, product costs, or non-positive order quantities found.
Date Coverage Validation	Sales data spans from 2005-07-01 to 2008-07-31 (37 months).
Referential Integrity Check	No orphan products, customers, territories, or dates detected.
Sales Quota Validation	Sales quota values appear complete and within reasonable ranges.

Conclusion

The dataset passed all major data quality checks. No significant issues were identified that would require data correction, removal, or imputation. The data is considered sufficiently complete, consistent, and reliable for further business analysis.

6. Baseline Business Metrics (Python)

Objective

Establish key business performance metrics from the *FactInternetSales* table to understand overall sales performance, profitability, customer activity, and order behavior before beginning detailed trend and segmentation analysis.

Analysis Performed

- Imported the FactInternetSales table into Python using pandas.
- Validated dataset size and structure.
- Calculated core business KPIs including:
  - Total Revenue
  - Total Product Cost
  - Total Profit
  - Profit Margin
  - Total Units Sold
  - Total Orders
  - Total Customers
  - Average Orders per Customer
  - Average Sales Amount
  - Maximum Sales Amount

Key Findings

Metric	Value
Total Revenue	$29,358,677.22
Total Product Cost	$17,277,793.58
Total Profit	$12,080,883.64
Profit Margin	41.15%
Total Units Sold	60,398
Total Orders	27,659
Total Customers	18,484
Average Orders per Customer	1.50
Average Sales Amount	$486.09
Largest Sale	$3,578.27

Summary

The dataset contains 60,398 sales records representing 27,659 unique orders from 18,484 customers. Revenue totaled approximately $29.36 million, generating $12.08 million in profit and an overall profit margin of 41.15%. Customers placed an average of 1.5 orders each, suggesting repeat purchasing exists but may offer opportunities for further growth. These baseline metrics provide a benchmark for subsequent analyses of sales trends, customer behavior, product performance, and profitability drivers.

7. Time Trend Analysis

Objective

Analyze how revenue, orders, customers, and average order value changed over time to identify overall business growth patterns and key performance trends.

Analysis Performed

- Calculated total revenue by year.
- Measured year-over-year revenue growth rates.
- Analyzed quarterly revenue trends.
- Counted unique orders by quarter.
- Counted unique customers by year.
- Calculated Average Order Value (AOV) by quarter.
- Compared quarterly performance across years while accounting for incomplete periods (2005 and 2008).

Key Findings

Metric	Finding
Revenue Growth	Revenue increased substantially from 2005 through 2008, though 2008 contains incomplete data and should not be directly compared with prior full years.
YoY Growth	Revenue grew by 99.9% from 2005–2006 and 49.9% from 2006–2007.

Quarterly Revenue	Revenue accelerated significantly during 2007 and the first half of 2008.
Order Volume	Quarterly orders increased from hundreds of orders in 2005–2006 to several thousand orders by 2007–2008.
Customer Growth	Unique customers increased from 1,013 in 2005 to 11,377 in 2008.
Average Order Value	AOV declined from over $3,200 per order in 2005–2006 to roughly $760–845 in early 2008.
Growth Driver	Revenue growth was primarily driven by increasing customer and order volume rather than higher spending per order.
Data Limitation	Q3 2008 contains incomplete data and was excluded from trend interpretation.
Seasonality	June,May, and December generated the highest revenue while September, August, and July were the weakest months.

Business Insight

The company experienced strong growth between 2005 and 2008, driven primarily by increases in customers and order volume. However, average order value declined significantly, suggesting growth came from a larger number of smaller purchases rather than higher spending per order. Revenue also showed seasonal patterns, with peak performance in late spring and December and weaker sales during the summer months. Further product-level analysis is needed to determine what products and customer behaviors drove these trends.

8. Product Analysis

Objective

Evaluate product, subcategory, and category performance to identify the main drivers of revenue and profitability, determine the most profitable products, and assess overall product portfolio performance.

Analysis Performed

- Calculated total revenue and profit by product category.
- Calculated profit margins by product category.
- Calculated total revenue and profit by product subcategory.
- Calculated profit margins by product subcategory.
- Identified top-performing products by revenue.
- Identified top-performing products by profit.
- Identified products with the highest profit margins.
- Attempted a product revenue trend analysis for discontinuation candidates, but the analysis was excluded due to data limitations and differing product launch periods.

Key Findings

Category Performance

- Bikes generated the highest revenue (~$28.5M) and profit (~$11.5M), making them the primary business driver.
- Accessories generated substantially lower revenue (~$0.70M) but achieved the highest profit margin (~62.6%).
- Clothing generated the lowest revenue (~$0.34M) and profit (~$0.14M).

Subcategory Performance

- Road Bikes were the most profitable subcategory, generating approximately $5.54M in profit.
- Mountain Bikes ranked second with approximately $4.51M in profit.
- Together, Road Bikes, Mountain Bikes, and Touring Bikes accounted for the overwhelming majority of total product profit.

- Several accessory subcategories (Helmets, Tires and Tubes, Bottles and Cages, Hydration Packs, Bike Racks, and Bike Stands) achieved profit margins of approximately 62.6%, significantly higher than bike subcategories.

Product Performance

- The highest-revenue products were dominated by Mountain-200 and Road-150 bike models.
- The highest-profit products were also concentrated among premium bike models, particularly Mountain-200 and Road-150 variants.
- High-margin accessory products achieved margins of approximately 62.6%, while many bike products generated margins ranging from approximately 36% to 46%.
- This indicates that bikes drive volume and profit dollars, while accessories provide stronger profitability on a percentage basis.

Product Trend Analysis

- A product discontinuation analysis was considered using quarterly sales trends.
- Due to varying product introduction dates, incomplete sales histories, and limited historical coverage, the results could have been misleading.
- The analysis was therefore excluded to maintain analytical reliability.

Summary

The product portfolio is heavily dependent on the **Bikes** category, which generates the vast majority of revenue and profit. **Road Bikes** and **Mountain Bikes** are the strongest-performing subcategories and represent the company's most important revenue drivers. While bike products generate the largest profit dollars, **Accessories** consistently achieve the highest profit margins, suggesting an opportunity to increase accessory sales through cross-selling and bundling strategies. Overall, the analysis indicates that maintaining strong bike sales while expanding high-margin accessory purchases could improve overall profitability.

9. Regional Performance Analysis

Objective

Evaluate regional sales performance to identify the highest-performing territories, assess profitability and efficiency across regions, compare customer value metrics, analyze product category performance by region, and examine regional revenue growth trends over time.

Analysis Performed

- Merged sales transactions with sales territory data using SalesTerritoryKey.
- Calculated total revenue and profit by region.
- Compared regional profitability using profit margin percentages.
- Calculated revenue per customer and profit per order for each region.
- Analyzed product category performance across regions.
- Examined revenue trends by region across multiple years.

Key Findings

- Australia generated the highest revenue and profit, making it the strongest-performing region overall.
- Southwest was the second-largest revenue-generating region and demonstrated strong growth over time.
- Profit margins were relatively consistent across all regions, ranging from approximately 41% to 45%, indicating no major regional cost-efficiency concerns.
- Australia achieved the highest revenue per customer and profit per order among major regions.
- Germany and the United Kingdom also demonstrated strong customer value metrics.
- Central, Northeast, and Southeast were excluded from customer-value comparisons due to extremely small customer and order counts.
- Bikes dominated sales across every major region, while Accessories and Clothing remained secondary revenue contributors.
- Product category rankings were highly consistent across regions, suggesting similar purchasing behavior across markets.
- Revenue generally increased across major regions between the complete years of 2006 and 2007.
- Australia remained the largest market throughout the available period, while Southwest showed particularly strong expansion.

Summary

Regional performance was driven primarily by Australia and Southwest, with strong contributions from the United Kingdom, Germany, and France. Profitability remained stable across territories, indicating consistent cost management. Customer value metrics highlighted Australia as the strongest market, while category performance showed remarkably similar purchasing patterns across regions. Overall, the analysis suggests that regional differences are driven more by market size and sales volume than by differences in product preferences or profitability.

10. Customer Analysis

Objective

Analyze customer behavior and demographic performance to identify which customer groups generate the highest revenue, profit, average order value, purchase frequency, and overall customer value. The goal was to move beyond individual customer names and focus on customer segments that can support better business decisions, targeting strategies, and future marketing recommendations.

Analysis Performed

- Loaded the customer dimension table and reviewed available customer demographic fields.
- Merged customer data with the existing sales-region dataset using `CustomerKey`.
- Verified that the customer-sales merge worked successfully.
- Analyzed top customers by total revenue, total profit, and number of orders.
- Added customer names, occupations, and income levels to make top-customer results more interpretable.
- Calculated average order value for top customers.
- Replaced the original top-customer question with a stronger customer-segment question because individual customer names were less actionable.
- Analyzed customer value by occupation using total revenue, total profit, unique customers, orders, revenue per customer, profit per customer, and purchase frequency.
- Created age groups from customer birth dates using the customer's age at the time of purchase.
- Analyzed customer performance by age group.
- Created income segments based on yearly income ranges.
- Analyzed customer value and profitability by income segment.
- Analyzed customer performance by gender to test whether gender was a meaningful driver of customer value.
- Compared income, age, occupation, and gender to identify which factors were most strongly associated with high-value customers.
- Replaced the repeat-purchase question with a stronger business question: "What factors are most strongly associated with high-value customers?"

Key Findings

Top Customer and Occupation Insights

- The top individual customers generated around $13K in total revenue each, with most placing five orders.
- Top customers were mostly from Professional and Management occupations, but individual customer names were less useful for business decision-making than segment-level analysis.
- Professional customers generated the highest total revenue and profit among occupation groups.
- Skilled Manual customers generated the second-highest total revenue, but this was mainly driven by customer volume rather than higher customer value.
- After normalizing by customer count, Professional customers generated the highest revenue per customer at approximately $1,795.
- Management customers ranked very close behind Professionals, with approximately $1,778 in revenue per customer.
- Manual customers were the weakest occupation segment, generating approximately $1,199 in revenue per customer.
- Occupation showed a meaningful difference in customer value, but the difference was weaker than income and age.

Age Group Insights

- Customer age was calculated based on the sale year instead of the current year, making the analysis more accurate for transactions between 2005 and 2008.
- The dataset did not contain teenage customers; the minimum customer age was 21.
- Age groups were created as 21–29, 30–39, 40–49, 50–59, and 60+.
- Customers aged 30–39 were the strongest age segment across multiple metrics.
- The 30–39 group generated the highest total revenue, highest total profit, highest revenue per customer, highest profit per customer, highest average order value, and highest purchase frequency.
- Customers aged 40–49 were the second strongest age segment.
- Customers aged 60+ generated the lowest revenue per customer, lowest profit per customer, lowest average order value, and lowest purchase frequency.
- The strongest customer age range was 30–49, suggesting that middle-aged working adults are the most valuable age demographic for AdventureWorks.

Income Segment Insights

- Income segments were created using yearly income ranges:

    - Low Income: under $30K
    - Lower-Middle: $30K–$60K
    - Middle: $60K–$90K
    - Upper-Middle: $90K–$120K
    - High Income: $120K+
- Lower-Middle and Middle income groups generated the highest total revenue because they had the largest customer bases.
- However, when normalized by customer count, High Income customers were the most valuable segment.
- High Income customers generated approximately $2,400 in revenue per customer and approximately $999 in profit per customer.
- Low Income customers generated approximately $1,215 in revenue per customer and approximately $496 in profit per customer.
- Revenue per customer and profit per customer increased consistently as income increased.
- This relationship showed that income was the strongest driver of customer value in the customer analysis.
- Higher-income customers were more valuable because they placed larger orders and purchased slightly more frequently.
- High Income customers had the highest average order value, approximately $1,441.
- Low Income customers had the lowest average order value, approximately $884.
- Purchase frequency also increased with income, from approximately 1.37 orders per Low Income customer to approximately 1.67 orders per High Income customer.

Gender Insights

- Female customers generated slightly more total revenue and profit than male customers, despite having slightly fewer customers.
- Female customers generated approximately $1,622 in revenue per customer and approximately $668 in profit per customer.
- Male customers generated approximately $1,556 in revenue per customer and approximately $640 in profit per customer.
- The difference between genders was small, around 4%.
- Gender appears to be a weak predictor of customer value compared with income and age.
- The analysis suggests that the business should not prioritize gender-based targeting as strongly as income-based or age-based targeting.

High-Value Customer Driver Insights

- Income was the strongest factor associated with high-value customers.
- Age was the second strongest factor, especially the 30–39 and 40–49 age groups.
- Occupation had a moderate relationship with customer value, with Professional and Management customers outperforming Manual customers.
- Gender had the weakest relationship with customer value.
- The strongest high-value customer profile was higher-income customers aged 30–49, especially those in Professional or Management occupations.
- The weakest customer value patterns appeared among Low Income customers, 60+ customers, Manual occupation customers, and customers with lower purchase frequency.

Summary

Customer analysis showed that the most valuable customers are not best understood by individual names, but by demographic and behavioral segments. Income was the strongest driver of customer value, with revenue per customer and profit per customer increasing consistently as income increased. Age was also highly important, with customers aged 30–39 and 40–49 producing the strongest revenue, profit, average order value, and purchase frequency. Occupation added useful context, especially for Professional and Management customers, but was less powerful than income and age. Gender had only a small impact and should not be treated as a major targeting factor. Overall, AdventureWorks should prioritize higher-income customers aged 30–49 for future marketing, premium product targeting, and retention strategies, while maintaining broader reach among middle-income customers because they contribute the largest total revenue volume.

11.Strategic Questions & Recommendations

Q01 — Which products should receive increased marketing and sales investment to maximize profit dollars?

Supporting evidence: Bikes generated ~$28.5M in revenue and ~$11.5M in profit, making them the dominant business driver. Road Bikes produced ~$5.54M in profit and Mountain Bikes ~$4.51M — together accounting for the overwhelming majority of total product profit. Mountain-200 and Road-150 are the top-performing individual products by both revenue and profit. Accessories achieved margins of ~62.6% versus 36–46% for most bike products. Clothing generated the lowest revenue (~$0.34M) and profit (~$0.14M) of any category.

Recommendations:

- Protect and grow Road Bikes and Mountain Bikes — they are the profit engine. Prioritize inventory depth and marketing budget for Mountain-200 and Road-150 lines, as these products directly determine whether the company hits its profit targets.
- Expand the accessory attach rate across the customer base. Accessories yield ~62.6% margins, nearly double the margin on a typical bike sale. Even a modest increase in accessories per order would materially lift overall portfolio profitability without additional cost infrastructure.
- Deprioritize or restructure the Clothing line. With the lowest revenue and profit in the portfolio, clothing currently absorbs resources — marketing spend, inventory carrying cost, shelf space — that could be redirected to higher-returning categories.

Q02 — How can the company improve profit margins without sacrificing revenue volume?

Supporting evidence: The blended profit margin sits at 41.15%, driven heavily by bike sales at lower individual margins. Accessories consistently deliver ~62.6% margins — significantly more efficient per dollar of sales — yet accessory revenue was only ~$0.70M, meaning their share of the portfolio is disproportionately small relative to their profitability potential. Revenue growth between 2005 and 2008 was driven by volume rather than higher per-order spending, and average order value declined from ~$3,200 in 2005–2006 to ~$760–845 by early 2008, compressing margin leverage.

Recommendations:

- Launch structured cross-sell and bundling programs pairing accessories with every bike purchase. Helmets, Tires and Tubes, Hydration Packs, and Bike Stands all clear 62.6% margins. A single accessory added to each bike transaction could shift the blended margin meaningfully upward.
- Introduce tiered pricing or premium configurations for high-demand models like Mountain-200 and Road-150. These products already command customer loyalty

— a modest price premium on top configurations would recover margin without requiring new customers or additional volume.
- Track AOV as a KPI alongside total revenue. Setting explicit AOV targets and incentivizing larger basket sizes through promotions, minimum-spend thresholds, or bundle discounts would counteract the declining spend-per-order trend.

Q03 — Which regions should be prioritized for targeted growth investment, and which require cost scrutiny?

Supporting evidence: Australia generated the highest revenue and profit of all regions and maintained the highest revenue per customer and profit per order. Southwest was the second-largest revenue region and showed particularly strong expansion between 2006 and 2007. UK and Germany demonstrated strong customer value metrics despite smaller total market sizes. Profit margins across all regions were tightly clustered at 41–45%, meaning regional differences are volume-driven, not margin-driven. Central, Northeast, and Southeast had extremely small customer and order counts.

Recommendations:

- Increase growth investment in Australia. As the company's largest and most profitable market with the highest revenue per customer, Australia is the safest place to deploy incremental budget. Expanding the customer base there leverages a market already proven to convert at high value.
- Accelerate Southwest expansion. Southwest's growth trajectory indicates market momentum that has not yet peaked. Adding sales coverage, localized promotions, or regional inventory depth now would capture growth before the market matures.
- Evaluate UK and Germany for premium-product campaigns. Both markets demonstrate strong customer value metrics — customers there are already spending at high per-order rates, so introducing higher-margin products or accessory bundles could produce outsized profitability gains relative to the required investment.
- Review cost allocation for Central, Northeast, and Southeast. These regions produce minimal revenue and orders. Any fixed costs — staffing, distribution, logistics — allocated to these territories should be reassessed and potentially redirected to high-performing regions.

Q04 — How concentrated is the company's revenue and profit risk, and what steps should management take to reduce exposure?

Supporting evidence: The Bikes category generates nearly all company revenue (~$28.5M of ~$29.4M total) and profit (~$11.5M of ~$12.1M total). Within Bikes, Road Bikes and Mountain Bikes account for the overwhelming share of subcategory profit. Australia alone is the largest market and contributes a substantial share of overall performance. Product category rankings are nearly identical across every major region, meaning there is no geographic diversification of product mix. Accessories and Clothing together represent a small fraction of total revenue and profit.

Recommendations:

- Formally acknowledge the concentration risk in board and management reporting. When a single category and a single market represent the majority of revenue and profit, any disruption — supply constraints, competitive entrants, demand shifts — creates outsized business impact. Awareness is the first step toward mitigation.
- Treat Accessories as a strategic diversification lever, not just an upsell opportunity. Growing accessory revenue from ~$0.70M to even $2–3M would meaningfully reduce category concentration while simultaneously improving overall margins — making this both a risk-reduction and a profitability play.
- Develop regional markets outside Australia more deliberately. Southwest shows growth momentum that, if sustained, could reduce the share of performance attributable to any single market. Structurally, having two or three large markets is less risky than one dominant one.

Q05 — Which customer segments should receive the most marketing and retention investment, and which are lower priority?

Supporting evidence: Income is the strongest predictor of customer value — revenue per customer increases consistently from ~$1,215 (Low Income) to ~$2,400 (High Income). High Income customers also have the highest AOV (~$1,441 vs. ~$884 for Low Income) and purchase frequency (1.67 vs. 1.37 orders). Customers aged 30–39 generated the highest total revenue, highest profit per customer, highest AOV, and highest purchase frequency of any age group. Customers aged 40–49 ranked second across the same metrics, making the 30–49 bracket as a whole the most valuable age demographic. Professional and Management occupation customers generated ~$1,795 and ~$1,778 in revenue per customer respectively. Gender had only a ~4% impact on customer value and is a weak targeting variable.

Recommendations:

- Concentrate retention and premium-product marketing on higher-income customers aged 30–49, particularly those in Professional and Management occupations. This segment generates the highest revenue per customer, highest profit per customer, and highest purchase frequency — they are the most cost-efficient cohort to market to.
- Design acquisition campaigns to attract middle-income customers while simultaneously nudging them toward higher-margin products through accessories bundling and tiered product recommendations. Middle-income customers generate the most total revenue in aggregate and should not be neglected.
- Avoid gender-based targeting as a primary axis. The ~4% gap between female and male revenue per customer is not large enough to justify separate product lines, pricing, or messaging strategies. Reallocate any gender-targeted budget toward income- or age-based segmentation, which yields much stronger differentiation.

Q06 — What resource reallocation across product categories would most improve overall business profitability?

Supporting evidence: Accessories generate only ~$0.70M in revenue but do so at a 62.6% margin — the most efficient category in the portfolio. Clothing generates the lowest revenue (~$0.34M) and the lowest profit (~$0.14M), contributing minimally while consuming resources. The company's blended margin of 41.15% is held down by the volume-weighted dominance of bikes at lower margin rates. Helmets, Tires and Tubes, Bottles and Cages, Hydration Packs, Bike Racks, and Bike Stands each individually achieve ~62.6% margins.

Recommendations:

- Redirect a portion of Clothing's marketing, inventory, and sales support budget toward Accessories. On a per-dollar-of-sales basis, accessories are nearly 50% more profitable than clothing. Eliminating or sharply reducing Clothing investment and reinvesting those resources into accessory range expansion and promotion would lift portfolio margins without requiring new customers.
- Expand the accessories product range and shelf presence. With only ~$0.70M in current accessory revenue and six subcategories each running at 62.6% margin, the business has significant headroom to grow this category through added SKUs, improved cross-sell visibility, and bundling with bike purchases.

- Evaluate whether Clothing should remain in the portfolio at all. If Clothing cannot be made meaningfully profitable through renegotiated costs or margin improvement, the resources tied to it would generate higher returns if redirected elsewhere.

Q07 — How should the company adjust its growth strategy given the pattern of rising volume but declining average order value?

Supporting evidence: Revenue grew 99.9% from 2005 to 2006 and 49.9% from 2006 to 2007 — strong headline numbers. However, AOV declined from ~$3,200 per order in 2005–2006 to ~$760–845 by early 2008, meaning growth came from smaller purchases rather than larger ones. Unique customers grew from 1,013 in 2005 to 11,377 in 2008, confirming volume — not value — drove topline growth. Revenue also showed seasonal patterns: June, May, and December peak while September, August, and July are the weakest months.

Recommendations:

- Introduce an AOV-improvement strategy to complement existing volume growth. The company is effective at acquiring customers but is not yet converting them to high-value buyers. Loyalty programs that reward repeat purchases, tiered discount structures that incentivize larger baskets, and bundle promotions timed to peak months (May, June, December) would address this gap directly.
- Capitalize on seasonal peaks with pre-planned inventory and marketing readiness. June, May, and December are confirmed high-revenue months. Allocating additional marketing spend and ensuring inventory availability during these windows would maximize revenue extraction from already-proven demand cycles.
- Monitor and set targets for profit per order, not just total revenue. Without this discipline, revenue growth can continue while overall profitability per unit of effort stagnates.

Q08 — Which regional markets contain the highest-value customers, and how should marketing budgets be allocated across them?

Supporting evidence: Australia achieved the highest revenue per customer and profit per order among all major regions. UK and Germany demonstrated strong customer value metrics despite smaller total market sizes. Southwest showed the strongest regional growth rate in the available period, suggesting an emerging high-value market. Product category performance is nearly identical across regions, meaning purchasing behavior does not vary significantly by geography. Profit margins across regions are tightly clustered at 41–45%, confirming that value differences stem from customer spending levels, not cost structures.

Recommendations:

- Allocate premium-product and high-margin accessory campaigns most heavily in Australia, UK, and Germany. These markets already demonstrate high per-customer value — additional investment there is amplified by a customer base already predisposed to spend more per transaction.
- Invest in customer acquisition in Southwest with a focus on quality over quantity. Southwest's growth momentum is proven, but the goal should be attracting customers in the high-income, 30–49 age bracket where possible, rather than purely maximizing volume — which risks replicating the AOV-decline pattern seen historically.
- Use consistent product messaging globally. Because category preferences are uniform across regions, the company does not need expensive regional product differentiation. A single product positioning strategy with localized language and pricing can be deployed across all markets, saving cost while maintaining consistency.

12. Dashboard Development

Objective

Develop interactive dashboards to communicate the project's key findings and provide stakeholders with an accessible way to explore financial, product, regional, and customer performance.

Dashboard Development (Completed)

An interactive dashboard was developed using Python and AI-assisted dashboard generation techniques. The dashboard consolidated the project's KPIs, findings, and recommendations into a visual reporting interface.

During dashboard development, multiple iterations were reviewed and refined to improve analytical accuracy, usability, and visual presentation. This process involved evaluating dashboard layouts, modifying and validating generated code, troubleshooting issues, and making adjustments to both Python and JavaScript components when necessary.

Skills Applied and Developed

- Dashboard design and layout planning
- KPI selection and business storytelling
- Data visualization best practices
- Python-based dashboard development
- Basic JavaScript code review and modification
- AI-assisted development workflows
- Dashboard testing, validation, and refinement
- Translating analytical findings into stakeholder-facing reports

Planned Power BI Implementation

A Power BI version of the dashboard is planned as an extension of the project. The objective is to recreate the dashboard using Power BI and compare the effectiveness, flexibility, and development experience across different dashboarding tools and technologies.

Note: The Power BI dashboard is currently under development and will be added to the project repository in a future update. The underlying analysis, findings, and recommendations presented in this project will remain unchanged.